K1 -- Summarize each step of the design process:

Defining the problem (Elements A - C)
Throughout the initial processes, our group first needed to find a problem that we would later research and solve. To this point, we have observed various issues reported by multiple teachers and stakeholders, ultimately deciding to undertake a project that would address a small, niche, yet important challenge facing hundreds of students at RM High School. The problem facing Forensic Science students occurred during a lab where students dropped blood on a piece of paper, altering the angle at which it was held, although this process proved to be extremely inaccurate and varied widely between students. After defining the issue, our group carried out extensive research on the problem at hand, looking for previous solutions, labeling possible stakeholders, and defining the specific constraints/requirements we would later need to work towards. After defining the various requirements and their order of importance, we were set up for success, having goals to achieve and criteria to follow.

Ideation, Selection, & Prototype design (Elements D - F)
With our criteria defined and the project centered around specific requirements outlined by the stakeholders, we spent hours crafting various designs and mechanisms that would possibly work to solve our problem. By using the stepladder method, the three of us formed our initial ideas, accumulated them into a single document, and built upon each idea. Using the particular advantages of each design, as well as omitting certain problems or downfalls of each design. While taking into account all the criteria provided, we collectively decided on a design that combined many of the positive aspects into one, functioning, aesthetically pleasing, and durable apparatus. After defining our particular solution, we researched the tools or, more importantly, the resources necessary to produce such a design. Finally, with all these factors taken into account, we set out a plan to carry out each day of the week, hopefully allotting the correct amount of time to each part of the process. This planning and preparation would, in turn, provide an outline and a “blueprint” for future construction.

The prototype building (Element G)
Initially, the building process was quite smooth; we measured and cut our wood to the desired lengths and widths with no problems. Our first problem arose when we realised the board could not sit entirely flat on the base because of the hinge. We quickly fixed this problem by planning to add another hole at the 0-degree mark to hold the board in place perfectly flat. We also ran into problems when drilling our holes and getting them to line up with the board. We fixed this by having multiple holes on the board so all the holes on the circular wood would line up properly. In the end, the prototype works as intended while also meeting all of our design requirements. Better woodworking skills would have allowed us to make it a little smoother to use, but the extensive planning still allowed us to design a successful prototype.

K2 -- What went well? What did not go well? Why?

The building and measuring out the wood went quite well. The function of the device worked almost perfectly but we did have to improvise for the holes that would hold the metal rod and stabilize the board on a certain degree. Originally, we had a plan to make two holes that would fit in all the holes carved for the different angles but we miscalculated and ended up making one connected hole that lined up perfectly. This quick thinking on our feet worked so well that it really tied this project together to make it both efficient and organized. Another thing we didn't consider was the clipboard and how we were going to attach that but luckily, Mr. Ackley provided us with the clipboard and we just needed to line it up and screw it in.

K3 -- Lessons learned – (reflection on K1 & K2)

1. It is essential for students to enjoy both the project and the problem they are undertaking. Two weeks is not a short time; therefore, to stay committed, on task, and centered on completing the requirements outlined by your stakeholder, as engineers, it is important to have some sort of interest and desire to do so. I recommend finding a problem that is complex and requires deep thought, instead of settling for a problem that requires 3 days of work, for example.
2. Although many group members were not here every day of the project due to exams and so on, it is important to communicate within your group throughout each day to ensure that no student is confused or outraged upon viewing the final product, which may not have been exaclty as one planned due to the constant "in-and-out" of many students during the construction process
3. Don't be afraid to ask for help; sure we as engineers are committed to solving these problems and figuring out solutions on our own, through the research and work we have put in leading up to the following step, although it is imperative that we accept help when needed, understanding that no one or few people may be able to complete a project solely by themselves. The teacher, other students, and adults are always open to giving suggestions, so to be afraid to ask!
4. A final lesson that our group collectively learned is that there will almost always be accidents or challenges that arise out of the blue, that, in many cases, were unable to be predicted beforehand. To this point, it is important that we as engineers develop the ability to counter these issues and persist through difficulties, ensuring that the project is completed no matter the random mistakes or problems that may occur throughout the planning and construction process.

What do you think designers need to consider or do based on your experience?

Based on our experience, our biggest takeaway is that extensive and proper planning is needed to produce a quality result. Moreover, it is imperative for all designers to take into account other viewpoints and possible iterations when considering the construction of their apparatus. In our case, each student brought there own skills and ideas towards the collective designing and creation of our mechanism, and without the use of varying ideas and a collection of iterations, the final design would likely be no where near as well thought out and "user-friendly" - displaying a clear consideration for other designers when formulating ideas for their given project or challenge. Finally, through our experiences in constructing this blood splatter apparatus, we recommend that other designers "relish the challenges" and persist through the various "road bumps" they may encounter. Problems are inevitable, but it is in the interest of the engineer to overcome those challenges and issues faced by inventing and ingenuing new and thoughtful ideas, therefore other designers must understand and consider this idea as well - to not give up after one or few things go "wrong" or were not exactly how you had planned it.